WHITEPAPER

PSOC™ Edge for Smart Glasses

Enabling cutting edge Smart Glasses with PSOC™ Edge MCU

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Abstract

In recent years, the Smart Glasses market has continued to expand rapidly with significant investments from major tech companies and strong public interest in the future of this technology. Smart Glasses have become popularized both for the practical value they bring today as well as the future potential of the technology – audio assistance for the hearing impaired, recording our most precious moments hands-free, providing real-time language translation and heads-up information, or interacting in completely immersive augmented experiences.

This whitepaper focuses on the emerging Smart Glasses market and outlines why PSOC™ Edge MCU is a well-suited platform for this application, delivering high-performance compute with AI/ML capabilities, leading power efficiency, and advanced audio/voice processing. In this whitepaper, we will start by walking through two typical Smart Glass architectures and corresponding design challenges. Then, we will explain the differentiated features which make PSOC™ Edge an ideal platform for Smart Glasses from the hardware definition and peripheral set to audio/voice middleware and AI/ML assets. Lastly, we will highlight additional key Infineon components which are proven in Smart Glasses and introduce the recommended PSOC™ Edge evaluation kit which can help a customer get started.

1 Smart Glasses is a growing market with a lot of excitement

Smart Glasses are an emerging category of head-worn wearable devices that seek to dramatically extend the functionality of everyday eyewear, adding tremendous value to the wearer's everyday activities. These devices can almost be thought of as lightweight computers – offering rich Human Machine Interfacing (HMI), real-time and capable compute, and connectivity to further expand device capabilities.

While Smart Glass implementations span the entire spectrum of capabilities and complexity, there are a few key features of Smart Glasses seen in the overwhelming majority of devices:

- **Capacitive Sensing On/Off Detection & Touch HMI:** Nearly every instantiation of Smart Glasses aims to be indistinguishable from everyday eyewear. In order to achieve this paradigm, OEMs developing Smart Glasses look to eliminate physical buttons. With this in mind, capacitive sensing is typically used to support elegant glasses on/off detection and touch-driven user inputs, typically on small touchpads integrated into the temple area.
- **Always-On Acoustic Activity Detection:** In order to support all day wear requirements, it is imperative to keep the Smart Glasses in the lowest operating power as much as possible, yet still ready to instantly interact with the user. Always-on Acoustic Activity Detection is the typical triggering mechanism to kick off the audio pipeline for Wake Word Detection, bringing the Smart Glasses into a higher performance domain for real-time usage and enabling a healthy balance in performance/power efficiency.
- **Rich Multi-Microphone Voice & Audio:** Voice and Audio are, bar none, the most broadly deployed HMI modality in the Smart Glasses ecosystem today. With many implementations featuring 5+ microphones, Smart Glasses today deploy Voice & Audio for use cases like Voice Assistant, Phone Call, Music Playing, Hearing Assistance, Speech-to-Text, Translation, and more!
- **Frictionless & Hands-Free Convenience:** Unlike Virtual Reality (VR) systems which often utilize dedicated Hand Controllers for human interaction, Smart Glasses deliver a hands-free way for the user to interact with the technology. In addition to touch HMI and Microphones, many Smart Glasses leverage cameras, Radar, and Time-of-Flight (ToF) to capture natural human inputs. Typically, Radar can be used for coarse interactions with virtual or augmented content (with 2D display overlays), whereas ToF enables rich 3D interaction with virtual elements.
- **Enhanced Connectivity:** Smart Glasses often leverage Wi-Fi, Bluetooth®, and BLE (BT Low Energy) to maximize their functionalities. The addition of connectivity enables connections to common peripherals like headphones, smart phones, or wrist watches and can also enable off-device compute for heavy duty compute workloads that would otherwise drain the battery in the Smart Glasses.
- **Cameras:** One of the first use cases deployed on Smart Glasses is RGB cameras. While traditionally used to effortlessly "capture the moment" with pictures, videos, or live streaming, cameras with AI to analyze the scene can be used to provide contextual awareness to the user.
- **Displays:** Arguably one of the most challenging technologies of the "extended reality" ecosystem, there are several prominent display technologies being featured in modern Smart Glasses – LCOS, uLED, and Laser Beam Scanning (LBS) to name a few. Even simple, single color displays add significant value to Smart Glasses due to their ability to provide notifications, directional overlay, language translation, and at the more extreme end – virtual content augmented onto and interacting with the world seamlessly.

1.1 Typical Architectures of Smart Glasses

Smart Glasses encompass a broad range of product implementations with diverse features and functionalities, addressing some or all of the above features. As a result, there are a wide array of customer architectures taking shape as well: Microprocessor (MPU) as the host, Microcontroller (MCU) as the host, MPU as host with an MCU co-processor, with and without displays, with and without cameras, with and without Wi-Fi, and Bluetooth® & BLE or BLE only connectivity implementations. Below we give two examples of Smart Glass architectures, and they are depicted in Figure 1 and Figure 2 respectively.

Figure 1 shows an example architecture of Smart Glasses with a Microcontroller (MCU) as the host and without a camera – commonly seen in Audio & AI-focused Smart Glasses. In this implementation, the host MCU is the primary compute SoC in this system and thus responsible for the totality of system management – sensor fusion, processing of voice and audio, managing the connectivity, and driving content to the display. Where possible, the host MCU will remain in the lowest operating power mode to be woken by several common triggers – most commonly touch HMI, capacitive sensing on/off detection or Acoustic Activity Detection (inbound Wake Word Detection) but sometimes a networking or IMU/motion event.

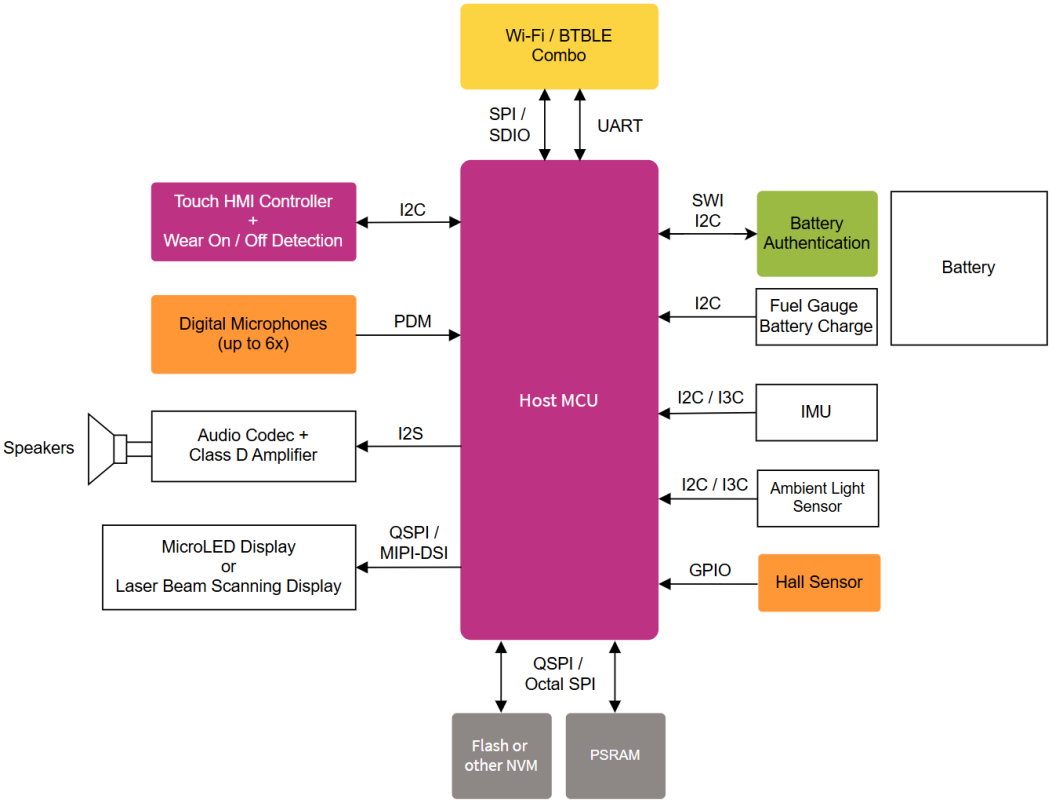


Figure 1 Architecture of Smart Glasses without camera

Figure 2 shows another example of architecture of Smart Glasses with a camera and a Microprocessor (MPU) as a host with an MCU co-processor. The addition of the camera is commonly used to support photo and video capture or live streaming but can also be used to support contextual awareness or human inputs such as hand gestures. While there are MCUs with video subsystems available today, many OEMs utilize an MPU due to the rich OS and easier interfacing with smart phones. However, there is a power penalty when utilizing an MPU so OEMs will often include an MCU to offset the power increase and to manage less demanding tasks (while MPU sleeps).

In this architecture, the MCU co-processor manages the majority of the tasks such as HMI, sensor fusion, Wake Word Detection and Voice Assistant capabilities while the MPU is reserved for peak performance tasks such as video recording or live video streaming, demanding both the compute performance and connectivity. Some OEMs have suggested that nearly 90% of their targeted Smart Glass features can be managed within the MCU co-processor. In order to further optimize at the system-level, OEMs will occasionally include features like a DSI Switch, allowing both the MPU or MCU to interface the display as needed.

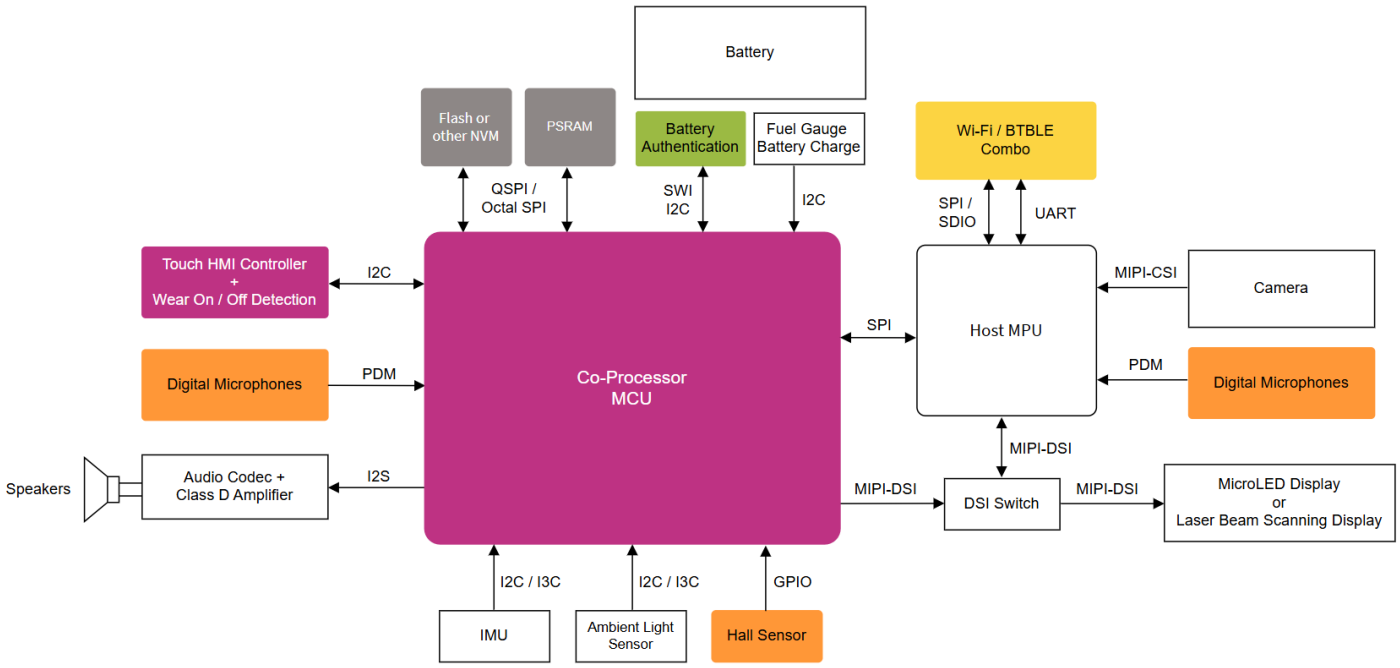


Figure 2 Architecture of Smart Glasses with a camera

1.2 Key challenges in designing Smart Glasses

In general, there are a few key challenges in designing Smart Glasses:

- **Small form factor** is a hard requirement, and it helps new generations of Smart Glasses look indistinguishable from traditional glasses and other everyday eyewear without the thick frames, physical buttons or obvious electronic components. This is usually achieved by using semiconductor components with smaller footprint and very tight system level integration. Small form factor also helps bring down the weight and make it easier to wear Smart Glasses for a longer period of time. In some cases, OEMs even prefer rectangularly shaped packages which are easier to fit into the temples of Smart Glasses!
- **Ultra-low power consumption** is especially important for Smart Glasses as their small form factor and light weight requirements dramatically limit the size and capacity of the battery. Low power consumption is crucial to extend battery life, allow longer use time before recharge, and enable continuous sensing even in sleep mode. The hardware and software need to be carefully tuned so that system level power consumption is balanced with the computing and AI/ML performance required for the targeted use cases.
- **AI/ML enabled functionalities** are getting popular, e.g., AI enabled voice assistant or ML based sensor fusion for improved accuracy and better situation awareness. This leads to the strong demand on the computing power and hardware acceleration for ML models running on Smart Glasses. Edge AI, without cloud services and remote computing assistance, has the benefits of reduced data communication with cloud, faster response time to events and the physical environment, a more interactive and immersive user experience.
- **Multi-microphone support** for better wake word detection and voice command accuracy. Previous implementations of voice assistants with a single microphone were not received well by users for its poor audio performance and usability when there is noise in the background. Multiple microphones also enable better Always-On listening for frictionless interactions with the user. At the same time, capable audio/voice processing is also important to provide robust and a smooth user experience on Smart Glasses.
- **Speaker diarization** is the process of identifying “who spoke when” in an audio clip. It is often done by segmenting the audio into regions based on the speaker’s identity. This important feature helps the Smart Glasses determine who speaks what at which point in a real-time conversation. Voice identification also helps the device to only respond to voice commands from the owner and does not trigger unnecessary actions in a crowded environment.
- **Wireless connectivity** is a key feature on Smart Glasses for the convenience and freedom of motion. Classic Bluetooth® as well as BLE are prominently featured for device synchronization with smart phones and stable audio streaming to earbuds. Many Smart Glasses also leverage Wi-Fi for companion smart phone use cases and to access the Internet, but some Smart Glasses directly integrate cellular 4G/5G network capabilities.

- **Embedded security** on Smart Glasses help protect personal and sensitive information for both the user and OEM. As users may wear Smart Glasses across many daily life scenarios and leave the device in public places like on a table of a coffee shop or bench, it is important to have appropriate data security and protection for user-specific information. Additionally, AI models have become high-value assets for OEMs and protecting them from market exposure is critical.
- **Limited suppliers** who have integrated hardware and software offering for Smart Glasses and AR experience. Usually there needs to be major coordination efforts across several key component vendors and packaging partners to make a new product. Hence, total solutions help tremendously for the lower system engineering cost and faster time-to-market.

2 PSOC™ Edge MCU platform for Next-Generation Smart Glasses Applications

2.1 PSOC™ Edge MCU family

PSOC™ Edge MCU is designed to provide the performance of an MPU and for next generation responsive compute and control applications, featuring hardware-assisted ML acceleration [1]. The PSOC™ Edge devices are based on high-performance Arm® Cortex®-M55 core [2], including Helium™ DSP support paired with Arm® Ethos™-U55 NPU [3], and Cortex™-M33 core [4] paired Infineon's own NPU, NNLite, with ultra-low power consumption.
Built in 22-nm node technology with embedded, non-volatile RRAM technology, this MCU family support Always-On sensing and response, which makes the devices ideal for advanced IoT devices such as Smart Home, Wearables, Robotics, and many more. This family of devices come with ample on-chip memories to support the most advanced applications and provide the scalability for increasingly sophisticated next-generation intelligent systems, while enabling rich software library for reuse and portability. See Figure 3 for the internal blocks of E84 MCU in this PSOC™ Edge family.

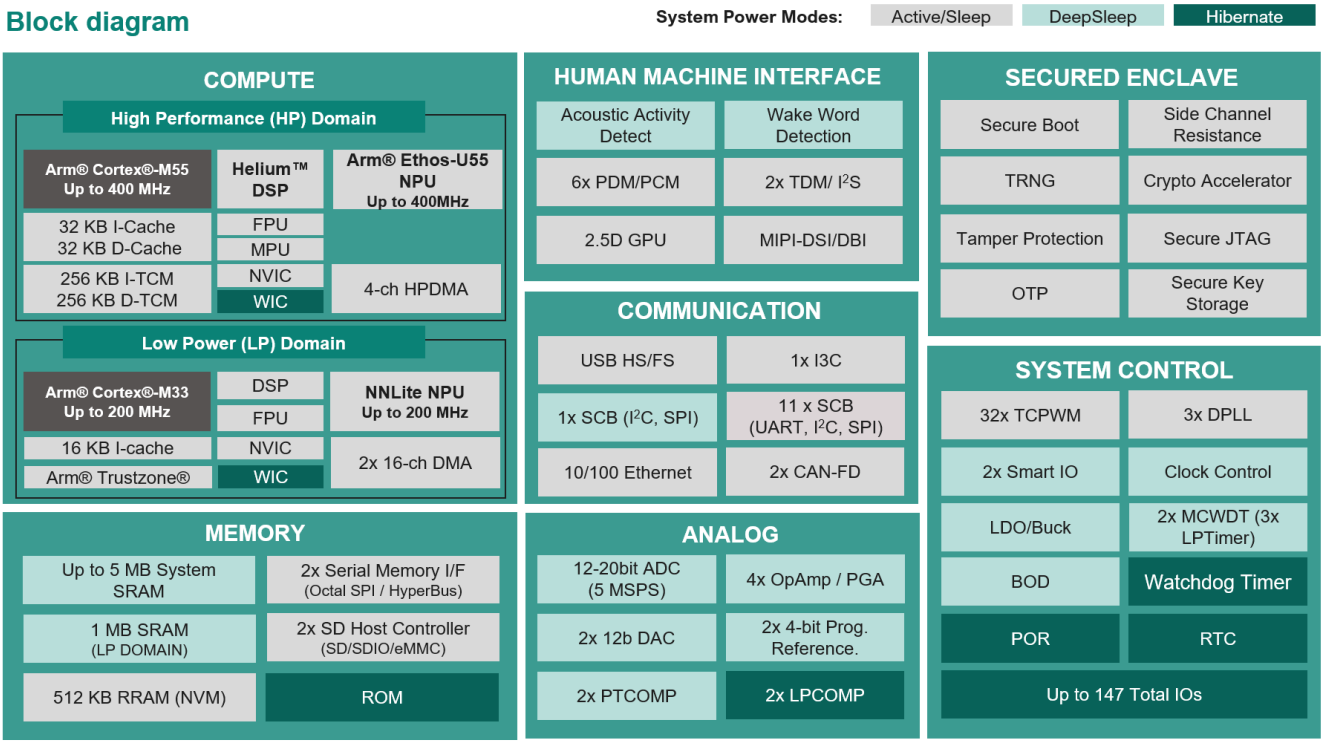


Figure 3 PSOC™ Edge E84 internal block diagram

While the PSOC™ Edge family is designed to deliver the performance of an MPU, it still offers the benefits of an MCU: E.g., 1) fast device boot time with RTOS, 2) a simple and flexible system design, 3) a compact PCB footprint and lower system BOM cost, and 4) lower system power consumption. The enhanced intelligence of the PSOC™ Edge MCU adds support for advanced graphics with 2.5D GPU and audio/vision-based applications to the extensive set of multi modal HMI. PSOC™ Edge family is also supported by strong ecosystem partners, comprehensive documentation, and the ModusToolbox™ software development environment [5], including the integration with best-in-class DEEPCRAFT™ Studio to build and run the AI/ML models [6]. This new series enables the easy migration of applications within the family as well as many upgrade paths from earlier designs.

There are also rich software supports for cloud SDK, connectivity drivers and audio/voice processing middleware as shown in Figure 4. There are out-of-box supports like Wi-Fi, Bluetooth and USB drivers, audio utilities, SW codec and HW codec. There are also ML based voice and audio solutions like Low Power Wake Word Detection, Voice Assistant, and multi-microphone Audio Enhancement with noise suppression, beamforming, and echo cancellation. These software modules can support many Smart Glasses use cases and enable quick product development and faster time-to-market.

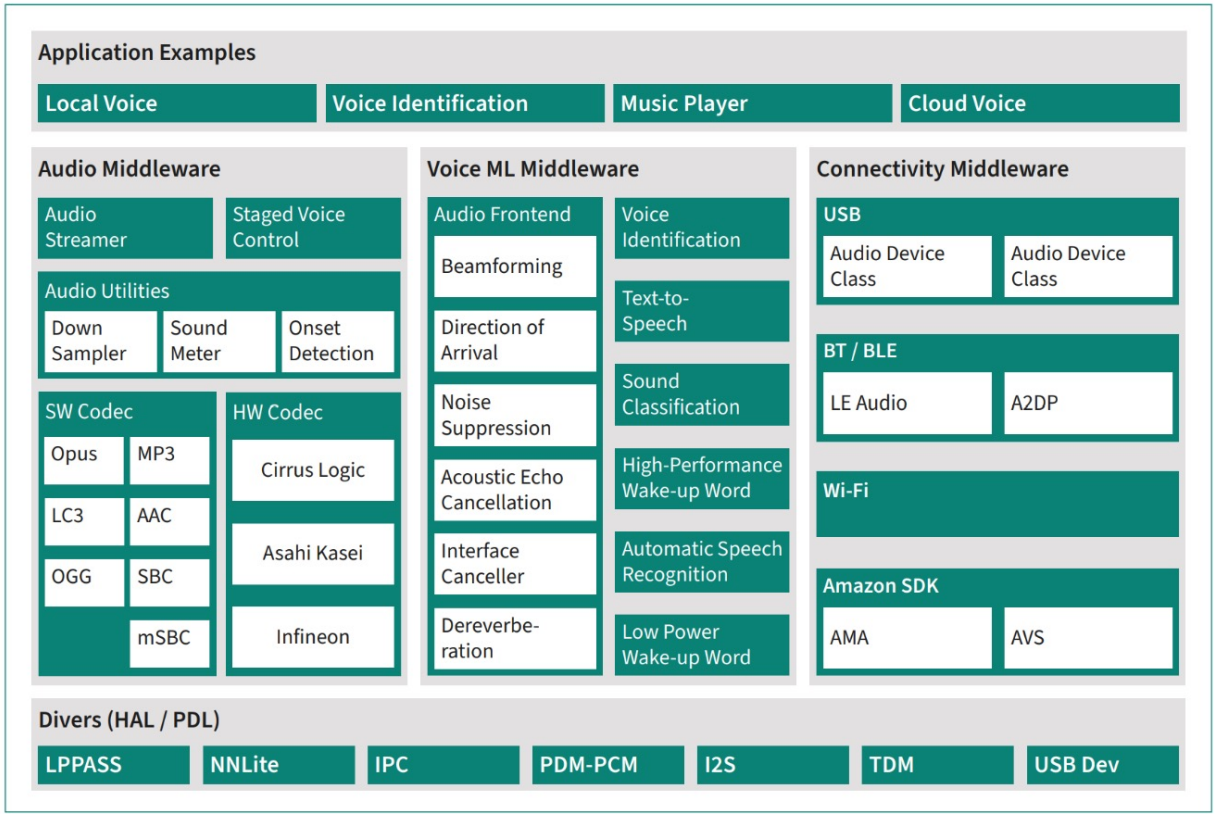


Figure 4 PSOC™ Edge device, audio and voice software support

2.2 Why is PSOC™ Edge well-suited for Smart Glasses embedded designs?

Specifically, there are a few key reasons why PSOC™ Edge MCU is a great platform to build Smart Glasses:

- **Multicore Architecture with Always-On, Low Power Compute, and High-Performance Domains:** PSOC™ Edge features a combination of Always-On, Low Power Compute, and High-Performance domains for end-to-end power and performance optimization. In Smart Glasses, the Always-On domain is typically used for Acoustic Activity Detection, pending an inbound Wake Word event. The Low Power Compute domain featuring M33 and NNLite NPU accelerates Sensor Fusion and manages the connectivity radio, waking the High-Performance Domain for peak performance tasks. Featuring an M55, Helium DSP, and U55 NPU, the High-Performance Domain supports rich Audio, Voice, and Compute Vision use cases while delivering fluid graphics for colorful HMI and animations on a display.
- **Autonomous analog for IOs and peripherals** can monitor sensor inputs independently and support Always-On sensing while both M33 and M55 subsystems stay in sleep mode to reduce power consumption. This autonomous analog circuitry forms the deep sleep domain in addition to the Low Power domain (with M33) and the High-Performance domain (with M55). Depending on the processing power required, the system can choose to wake up either the Low Power domain M33 or the High-Performance domain M55 (or both cores) for further sensor data processing. See Figure 5 for the three domains of E84: High-Performance domain (with M55), low power domain (with M33), and deep sleep domain (with autonomous analog).

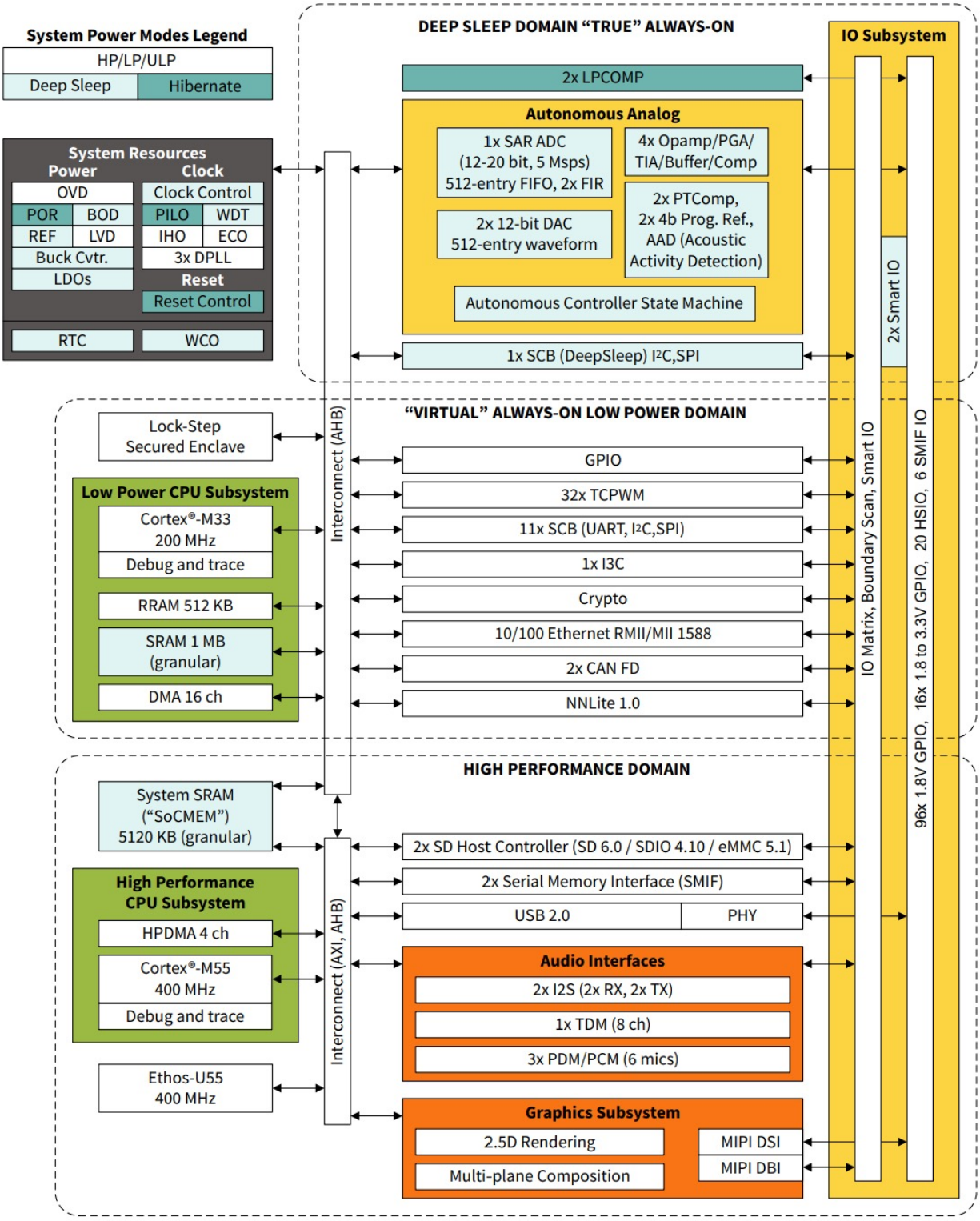


Figure 5 PSOC™ Edge E84's High-Performance domain, Low Power domain and Always-On domain

– **Low power consumption and power efficient compute** because of its advanced technology node process and efficient MCU internal architecture considerations offering low-power domains to allow Always-On (AON) operations and turn on the high-performance domain when relevant, PSOC™ Edge demonstrates leading power and energy efficiency when compared with applicable competitors.

– **AI/ML acceleration** in Smart Glasses provide key functional differentiation and power optimization. Fundamentally, PSOC™ Edge MCU offers Smart Glasses high AI/ML performance for Audio, Voice, and Computer Vision AI along with Always-On, low power sensing to support all day wear. While M55 alone with its Helium™ vector extensions & DSP can run ML models, M55 + U55 delivers much better ML performance compared to any previous generations of MCUs. Table 1 shows a comparison of acoustic event detection, and M55 + U55 achieves 109x faster inference speed compared to M4. Table 2 shows a similar comparison of ML based sensor fusion, and the inference speed running on M55 + U55 compared to M4 is 82x faster. Overall, M55 + U55 can achieve higher performance while reducing cycle counts, which help reduce the power consumption too. Please refer to this whitepaper [10] for more information.

Table 1 Acoustic event detection ML comparison

	Core Frequency (MHz)	Inference Time (ms)	Inference Time Improvement	Cycle count
M4 – Int8	150	30.459	baseline	4,568,850
M33 – Int8	200	25.999	1.17x	5,199,800
M55 + U55 – Int8	400	0.280	**109x**	112,000

Table 2 Sensor fusion based ML comparison

Supplier	Product & Core(s)	Frequency (MHz)	AudioMark Score	Score/MHz	Power (mW)	Power/Score (uW/Score)
Competitor 1	M55 + U55	400	5760.0	14.4	122	21.2
Competitor 2	M55 only	400	3540.8	8.9	98.3	27.8
Competitor 3	M55 only	96	326.8	3.4	11.6	35.5
		250	875.1	3.5	15.4	17.6
Infineon	**PSOC™ Edge E84 M55 + U55**	400	6247.8	15.6	81.9	13.1

– **Advanced audio/voice processing.** In our benchmark, M55 + U55 with the same software stack can increase up to 30x of the ML performance on a M55 only platform. For example, PSOC™ Edge with M55 + U55 enables Speech-to-Intent to create best-in-class voice commands performance with 95% of M55 still available and 3.5x increase of available M55 bandwidth (as in Full Audio Benchmark). This leaves significant computing and memory bandwidth resources available for other customer applications running on the PSOC™ Edge MCU. See Table 3 for the AudioMark score comparison of PSOC™ Edge E84 against several competitor MCUs. PSOC™ Edge shows significant advantages of higher performance per MHz and lower power consumption per score vs. competition with similar cores (M55 only or M55+U55).

Table 3 AudioMark Score comparison

Supplier	Product & Core(s)	Frequency (MHz)	AudioMark Score	Score/MHz	Power (mW)	Power/Score (uW/Score)
Competitor 1	M55 + U55	400	5760.0	14.4	122	21.2
Competitor 2	M55 only	400	3540.8	8.9	98.3	27.8
Competitor 3	M55 only	96	326.8	3.4	11.6	35.5
		250	875.1	3.5	15.4	17.6
Infineon	**PSOC™ Edge E84 M55 + U55**	400	6247.8	15.6	81.9	13.1

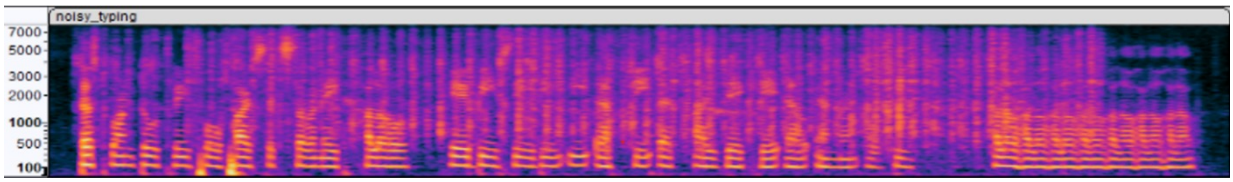
- **Ample memory support for high-performance and low power consumption.** PSOC™ Edge family supports up to 5MB SRAM, allowing multiple workloads to coexist simultaneously. There are two Serial Memory Interface (SMIF) with 32 KB cache each, supporting external memory up to 128 MB. There is 512KB RRAM as on-chip NVM for the storage of application configuration data, and also for small size code base to allow M33 subsystem to wake up quickly and execute the code before going back to sleep. This RRAM technology draws much lower current comparing with the XIP (Execution in Place) approach which execute the code directly from external Flash.
- **Robust peripheral supports to accommodate Smart Glasses system requirements.** PSOC™ Edge MCU has 2.5D GPU with interface MIPI-DSI and DBI. For sensors and peripherals, there are up to 12 Serial Configurable Blocks (SCBs) that can be configured to function as UART, I2C or SPI. During deep sleep, 1 SCB can be enabled to support continuous sensing via I2C and SPI. There is also a I3C controller for high speed peripherals with 25 MHz Double Data Rate (DDR) support. There are two TDMs with 8 channels each, supporting I2S full-duplex and half-duplex modes speaker output. For audio and voice processing, there are six channels of PDM/PCM to accept inputs from multiple microphones (up to 6x).
- **Lower BOM cost and simpler system design.** PSOC™ Edge MCU has high computing performance and AI/ML acceleration similar to MPU. High-performance MPU costs a lot more for total system components, including MPU, PMIC, eMMC, DRAM, and the PCB assembly. An MPU system usually draws much higher current running a task, compared with that in M33 or M55 subsystems in addition of the external components to be added i.e. memory controller, power management integrated circuit (PMIC) and external memories. PSOC™ Edge MCU family can significantly reduce the BOM cost, aggregated power consumption and total system footprint.
- **PSA Certified L3/L4 iSE** supports the highest level of hardware security with a Secure Enclave [9]. This level of security also incorporates protections for high value assets like a customer's AI models.

2.3 Rich AI/ML software supports on PSOC™ Edge MCU

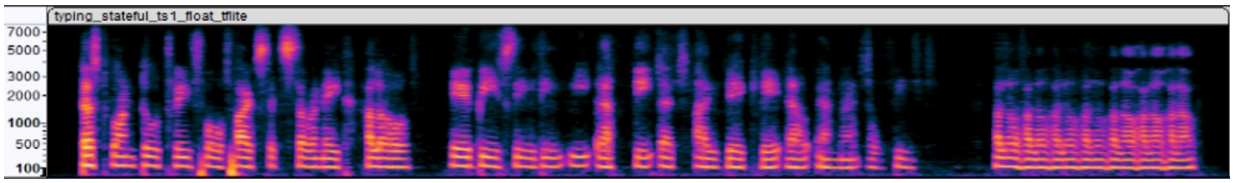
On Smart Glasses, AI/ML acceleration and the learning capabilities have become a key feature differentiator for the intelligence and convenience provided to the user. Below we give a few examples of AI/ML software supports from Infineon that are already proven to work well with PSOC™ Edge MCU.

- **DEEPCRAFT™ Voice Assistant** enables developers to easily add high quality voice features (commands, keywords, wake word detection, etc.) in the application. This solution provides low power, Always-On voice processing chain accelerated on PSOC™ Edge. Speech-to-Intent leveraging ML to create the best-in-class voice command performance with high accuracy and low false detects. This solution is also versatile for numerical digits recognition. There is an easy-to-use cloud tool to create the wake word and the set of voice commands without actual voice sample data. On PSOC™ Edge MCU, this wake word detection consumes less than 1 mW out-of-box while the Voice Assistant (with 20 Commands) consumes 7 mW.
- **DEEPCRAFT™ Audio Enhancement** can preprocess incoming speech data with background noise to have a clearer speech data, e.g., before being fed into voice command recognition. Specifically, Noise Suppression and Acoustic Echo Cancellation can remove the majority of environmental noises while keeping the fidelity of original voice from the user. Audio Scene Analysis can profile various acoustic artifacts to only extract desired ones. Beamformer with

multiple microphones can improve the SNR ratio of the voice source. Figure 7 shows an example of noisy speech sample and the processed output after applying this voice enhancement. On PSOC™ Edge MCU, this audio enhancement consumes only 22mW with all the above features. Please refer to this whitepaper [11] for more information.



(a) Noisy speech sample without any voice enhancement (before)



(b) DEEPCRAFT ™ Audio Enhancement applied on the speech sample (after)

Figure 6 DEEPCRAFT™ Audio Enhancement on speech with background noise: (a) before vs. (b) after

- **DEEPCRAFT™ Face ID (early access)** works with an RGB camera feed or a video file as input. This solution uses state-of-the-art cascaded neural network architecture to deliver exceptional accuracy (>96%) with extremely low false positives (one in a million). This Face ID also supports multi-face detection in the same video frame and tracks multiple faces simultaneously. The enrollment process is streamlined: it only takes a few minutes to capture a new user's face from different angles and then load the model into the embedded system. This solution is also designed to handle lots of enrollments, with testing up to 1500 unique faces.
- **DEEPCRAFT™ Object Segmentation (early access)** also works with a video input and can segment known objects in the video frame. This solution runs on PSOC™ Edge MCU with very high accuracy, maximizing Mean Average Precision (MAP) mask. It is highly customizable to train the model to segment any object in the frame for the target use cases. The solution can also segment and track multiple objects in the video simultaneously, enabling seamless video monitoring, object tracking and scene analysis. This also allows real-time video effects, superimposed on top of the segmented objects, for user privacy filtering and immersive augmented reality experiences.
- **DEEPCRAFT™ Ready Models** to enable low power, Always-On Acoustic Activity Detection (AAD) like baby cry, coughing, snoring, siren, alarm, and the direction of sound arrival.
- **DEEPCRAFT™ Gesture Detection** works with Infineon's 60GHz mmWave XENSIV™ Radars. This solution can detect a selected set of gestures based on the Radar signal emitted and then reflected by the user. The ML model has been optimized for common scenarios, providing smooth and effective HMI. Running on PSOC™ Edge MCU, this gesture detection model can classify gestures with high accuracy at the distance of 10 cm to 70 cm from Radar.

2.4 Other Infineon product offerings for Smart Glasses

Besides the host MCU, Infineon provides many key components for Smart Glasses that have been both proven in market and proven to work well with PSOC™ Edge MCU family. Infineon offers high integration options and has a lot of experience working with module partners and supports wide portfolio for flexible integration options.

- **AIROC™ Wi-Fi and Bluetooth® connectivity chips** offer best-in-class RF performance & optimized power for Smart Glasses use cases. For example, Infineon's Ultra Low Power Wi-Fi/BT combo radios like CYW43022 and CYW5551x have been used in Smart Glasses in market today. CYW5551x specifically is a combo chip with 1x1 Wi-Fi 6/6E and Bluetooth® 6.0 (both classic BT and BLE). It is power-optimized for Smart Glasses and supports live video streaming and fast file transfer with its long range and industry-leading RF performance. Its BLE also features HID enhancements to achieve very low HID latency around 1ms.

- **PSOC™ 4000T or 4100T Plus with the 5th Gen CAPSENSE™.** Smart Glasses typically use CAPSENSE™ to integrate On/Off Detection & Touch HMI functionality. Infineon's PSOC™4000T is proven in numerous Smart Glass designs today where customers value the ultra-low power scanning, high SNR, and robustness to hair, water, and sweat. For example, the device can scan-for-proximity/touch without core activation (< 6 uA) while the high SNR maximizes the detection distance and HMI responsiveness.
- **XENSIV™ MEMS microphones** have digital versions and analog versions. For example, IM66D130M MEMS is an ultra low power digital PDM microphone with ultra compact package. There are usually multiple, 2x to 5x, microphones per design to better support wake word detection and voice command. Other value propositions include low IDD 550/175, high Acoustic Overload Point (AOP) at 130 dB Sound Pressure Level (SPL) and 66 dB SNR. These microphones also enable ambient noise suppression and acoustic event detections, which further enhance the features of Smart Glasses. Algorithms and ready-to-use AI/ML models can be offered to customers for quick feature development and productization.
- **XENSIV™ Time-of-Flight camera.** Infineon's ToF camera enables 6DOF tracking and captures a higher level of details during depth and environmental sensing. ToF technology on Smart Glasses & AR thus allows real-time interactions with augmented 3D content projected on top of the user's environment with speed and precision. Infineon REAL3™ 3D ToF imager IRS2877C with VCSEL driver IRS9102C have been proven publicly in AR glasses.
- **60GHz mmWave XENSIV ™ Radar.** 60GHz Radar provides finer time-space resolution than the usual 24GHz Radar, allowing better space sensing and gesture control HMI with high accuracy on Smart Glasses. Infineon's BGT60TR13C enables interactions with 2D augmented content (e.g., swiping displays left, right, up or down). It also consumes much lower power compared with a ToF camera (however, a ToF camera in general provides a higher space granularity than a Radar). Radar gesture ML model also gains up to 10x inference speed running on PSOC™ Edge compared with running on M33. Infineon also offers complete Blackbox solution with both Radar hardware and processing algorithm.
- **External memory solutions.** ULTREON™ RRAM is NVM with the lowest power consumption and the smallest footprint. This RRAM is more durable than Flash and Byte writable with multi-bank for simultaneous reading and writing. Infineon also offers external memory solutions, e.g., SEMPER™ Nano NOR Flash and HYPERRAM ™ PSRAM, balancing R/W performance and the operating energy consumption.

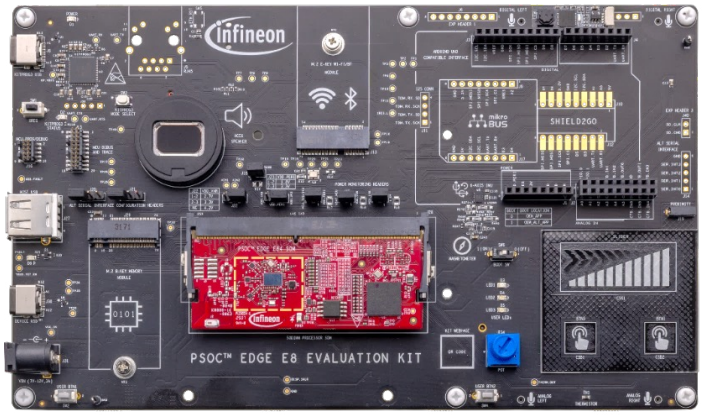
3 How to get started with PSOC™ Edge MCU?

3.1 Hardware enablement with PSOC™ Edge Evaluation Kit

The PSOC™ Edge Evaluation kit (EVK) enables a customer to evaluate and develop IoT applications using the E84 MCU along with on-board multimedia, Machine Learning (ML), and connectivity features like custom MIPI-DSI display, audio interfaces, and modules with AIROC™ Wi-Fi and Bluetooth® connectivity chip.

As part of the PSOC™ Edge family, this E84 MCU is a true programmable embedded system-on-chip with dual cores, integrating a 400MHz M55 as the primary application processor, a 200MHz M33 that supports low-power operations, and a 400MHz U55 NPU, NNLite, graphics and audio block, DSP capability, security enclave with crypto accelerators and protection units, high-performance memory expansion capability (QSPI, Octal SPI, and Octal HYPERRAM™), low-power autonomous analog subsystem with high-performance ADC and low-power comparators, communication channels, programmable analog and digital blocks that allow higher flexibility, in-field tuning of the design, and faster time-to-market.

This EVK carries a E84 MCU on a SODIMM-based, detachable SOM (System on Module) board, connected to the main baseboard. The MCU SOM board also has 128 Mb of QSPI flash, 1Gb of Octal flash, 128 Mb of Octal RAM, PSOC™ 4000T as CAPSENSE™ co-processor, and on-board AIROC™ Wi-Fi and Bluetooth® connectivity chip, CYW55513. See Figure 8 for the SOM board and how it fits into the main EVK baseboard. The final EVK assembly also includes a 4.3 inch display and a RGB camera connected via USB.



(a) The main EVK baseboard (dark grey) and the SOM board (red)

(b) The EVK assembly with 4.3 inch display on top and a RGB camera connected via USB

Figure 7 PSOC™ Edge E84 main board (dark grey), SOM board (red) and the EVK assembly with 4.3 inch display

The baseboard features an on-board programmer/debugger (KitProg3), ETM/JTAG/SWD debug headers, both analog and digital microphones, an Raspberry-Pi compatible MIPI-DSI connector, USB host Type-A and USB device Type-C connectors, a headphone jack, a speaker, an RJ45 Ethernet connector, an M.2 (B-key) memory interface and an M.2 (E-key) radio interface, Infineon's Shield2Go interface, a 6-Axis IMU sensor, a 3-axis magnetometer, mikroBUS compatible headers, a microSD card holder, CAPSENSE™ buttons and slider, indicator LEDs, and a few physical buttons. The E84 MCU power domain supports 2.7 V, 3.3 V, and 4.2 V operating voltages, and the peripheral power domain supports 1.8 V and 3.3 V operating voltages. Figure 9 shows the block diagram and major components of this E84 EVK.

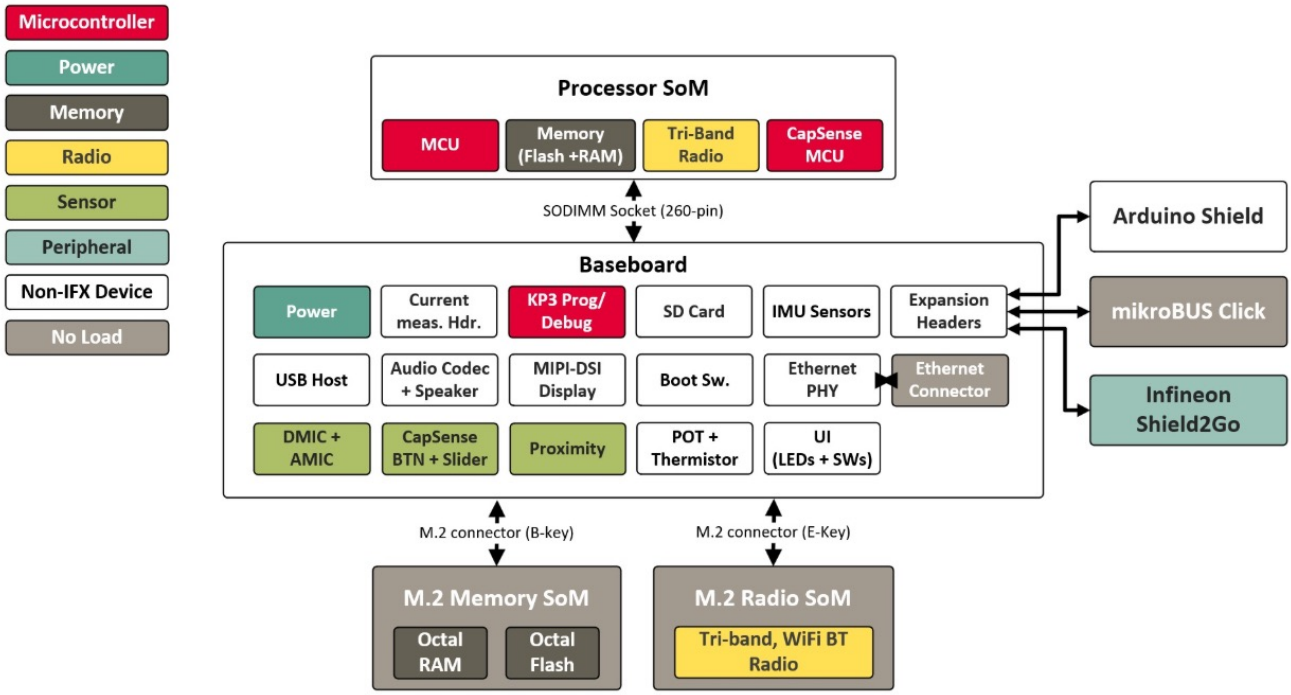


Figure 8 PSOC™ Edge Evaluation Kit Block Diagram

This PSOC™ Edge EVK will be available for purchase on Infineon website and major distribution channels. To have early access, please contact Infineon sales now. This EVK is supported by complete documentation of schematics, BOM list, pinout connection, software examples to start building a new product. At the moment, the PSOC™ Edge E84 Early Access Pack (EAP) has all software support and demo code examples, including LVGL Music Player Demo, LVGL Smart Watch Demo, GPU vs no-GPU rendering performance, Run-Length Adaptive Dithering (RLAD), Single vs Double Buffering, and VGLite 2D vector drawing acceleration. An updated SOM board is in planning to include an FCC certified module of the same AIROC™ Wi-Fi / Bluetooth connectivity chip. This updated SOM allows the easy adoption in customer products without the need to go through FCC certification.

3.2 Software development and supports on PSOC™ Edge

Infineon's ModusToolbox™ provides a comprehensive software development experience for the latest generation of MCUs like PSOC™ Edge family. It is more than an Integrated Development Environment (IDE) or software SDK delivery mechanism. Figure 10 shows the typical software development workflow on ModusToolbox™. The development tools, supporting Windows, Linux, and macOS, are available as a tool installation package from Infineon Developer Center. These desktop applications enable the creation of new embedded applications, managing software components, configuring device peripherals and middleware, and embedded development tools for compiling, programming, and debugging.

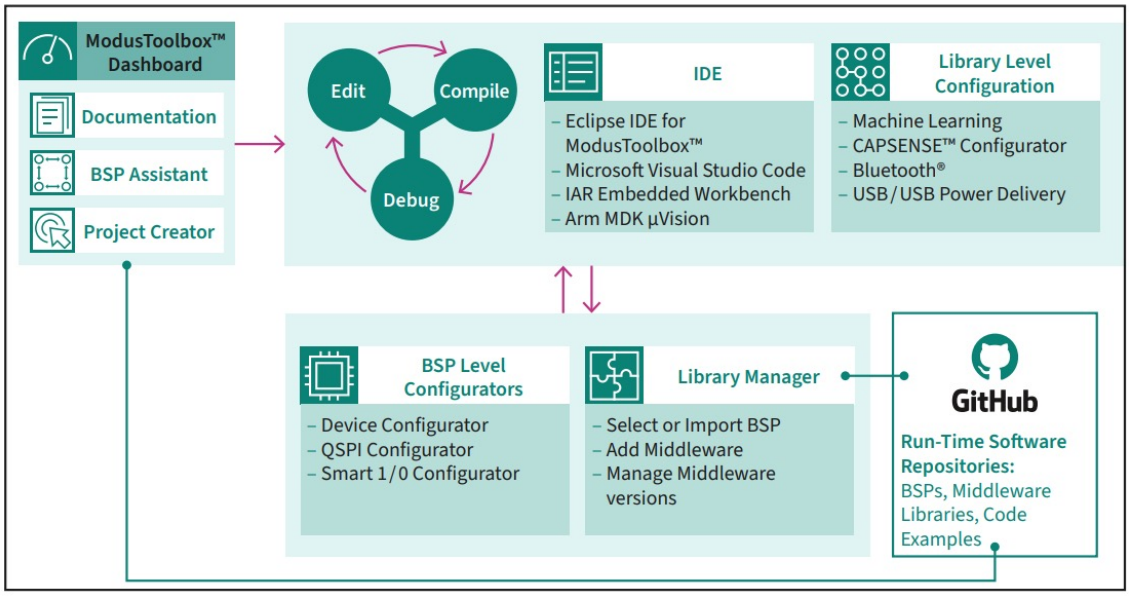


Figure 9 ModusToolbox™ software development workflow

ModusToolbox™ features a make build automation tool with structured BSP targets. This build system is accessible through a command-line interface or integrated within Microsoft Visual Studio Code or the included Eclipse IDE for ModusToolbox™, leveraging the Arm GNU Toolchain GCC. Additionally, IAR Embedded Workbench and Arm MDK with µVision IDE are supported out-of-the-box, making use of their respective compilers and IDE development workflows. ModusToolbox™ encompasses the entire Edit / Compile / Debug workflow and offers additional support related to the composition and configuration of an embedded project.

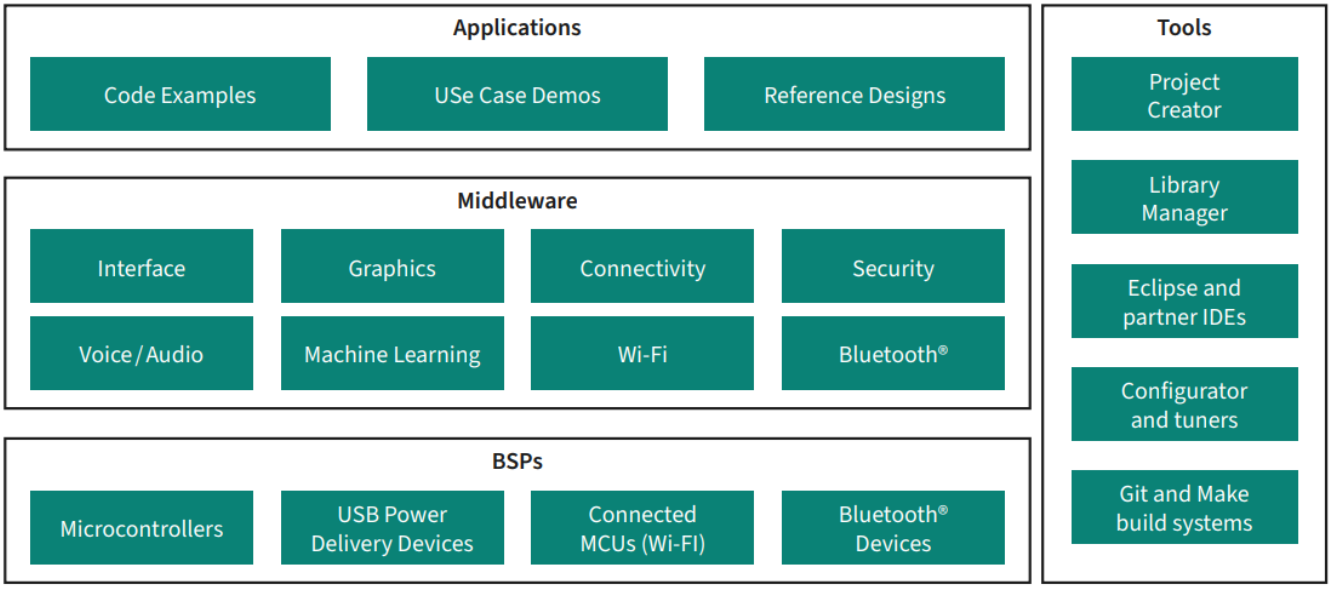


Figure 10 ModusToolbox™ software development resources

ModusToolbox™ interfaces directly with the available run-time software repositories, providing easy access to the latest development resources. Run-time software include middleware, BSPs (board support packages), device drivers and code examples, and they are provided via an extensive collection of GitHub-hosted repositories [7]. This allows both Infineon and partners to rapidly respond to and distribute updated code releases.

See Figure 11 for the software development resources available on ModusToolbox™. Tools like Project Creator and Library Manager interface directly with online repositories, facilitating the integration of the latest ModusToolbox™ resources. There are mainly three types of software supports:

- **BSPs** (Board Support Packages) are the foundation of the software support in ModusToolbox™. BSPs include packages that encompass device support for a wide range of supported microcontrollers, programmable power delivery controllers, Wi-Fi devices, and Bluetooth® SoCs. In addition to the device level support, these BSP targets provide excellent starting references for development kits from Infineon and partners supporting ModusToolbox™.
- **Middleware:** The next layer of software resources available within the ModusToolbox™ ecosystem are the rich set of middleware repositories including, but not limited to HMI, Graphics, Connectivity, Security, processing of Voice / Audio, AI and Machine Learning, Wi-Fi and Bluetooth® drivers.
- **Application code examples:** Code examples and reference applications complete the list of available resources in ModusToolbox™, providing sample projects ranging from simple peripheral use case examples to complex IoT connectivity references that can be used as starting points for application development.

There is also a strong community on Infineon developer forum [8] to ask questions and share experiences. Please refer to this whitepaper [12] for more information about ModusToolbox™ Software.

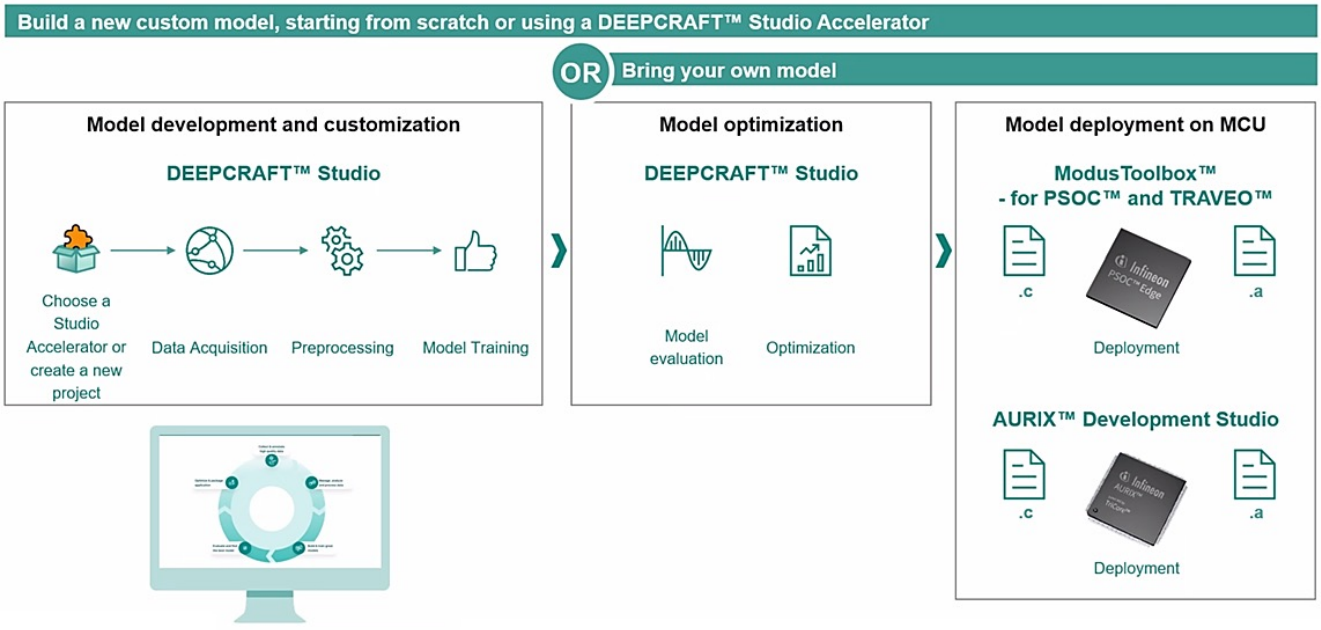


Figure 11 Figure 11 DEEPCRAFT™ Studio to support different ways of training a new ML model or bring your own model

Finally, DEEPCRAFT™ Studio provides easy and intuitive ways to build and run AI/ML models. DEEPCRAFT™ Studio is fully integrated with ModusToolbox™. In this studio, a user can select to “build a new custom model” by preparing the dataset, preprocessing the data and extracting features to train a new ML model. The studio provides a convenient UI to visualize and tag the data sequence for supervised learning. A user can also select to “bring your own model” by taking an existing ML model and optimizing it against the target MCU platform.

Figure 12 illustrates these two options to “build a new custom model” or “bring your own model” in DEEPCRAFT™ Studio. In both cases, the ML model can then be validated with a testing dataset and deployed to run on the MCU through the integrated ModusToolbox™. DEEPCRAFT™ Studio also provides sample starter projects and ready-to-use models for faster product development. If the MCU platform is from Infineon, including PSOC™ Edge family, this DEEPCRAFT™ Studio and the ready ML models are completely free to use, which is another great reason to build your applications on PSOC™ Edge platform.

4 Conclusion

PSOC™ Edge MCUs redefine what's possible in next-generation smart glasses by combining high-performance compute, advanced analog integration, and efficient AI processing in a compact, power-optimized architecture. With its built-in Edge AI capabilities, low-latency, secure connectivity as well as its innovative approach with a low-power domain for AON operations and a high-performance domain to process the demanding operations required in the new edge AI era, PSOC™ Edge empowers designers to deliver immersive, context-aware experiences without compromising battery life or form factor.

By leveraging Infineon's comprehensive development ecosystem—including ModusToolbox™ and DEEPCRAFT™ AI Suite—developers can rapidly prototype, train, and deploy AI models tailored to smart glasses use cases such as gesture recognition, visual assistance, real-time translation, and augmented reality overlays. This seamless hardware-software integration shortens development cycles while ensuring reliable, scalable performance.

Ultimately, PSOC™ Edge enables the next generation of wearable intelligence—smart glasses that are not only lighter and smarter, but truly aware of their environment and the user's intent. With its blend of performance, efficiency, and AI readiness, PSOC™ Edge stands as the ideal foundation for bringing cutting-edge, human-centric wearable innovation to life.

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Published by
Infineon Technologies AG
Am Campeon 1-15, 85579 Neubiberg
Germany

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Public

Version: V1.0_EN
Date: 10/2025

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